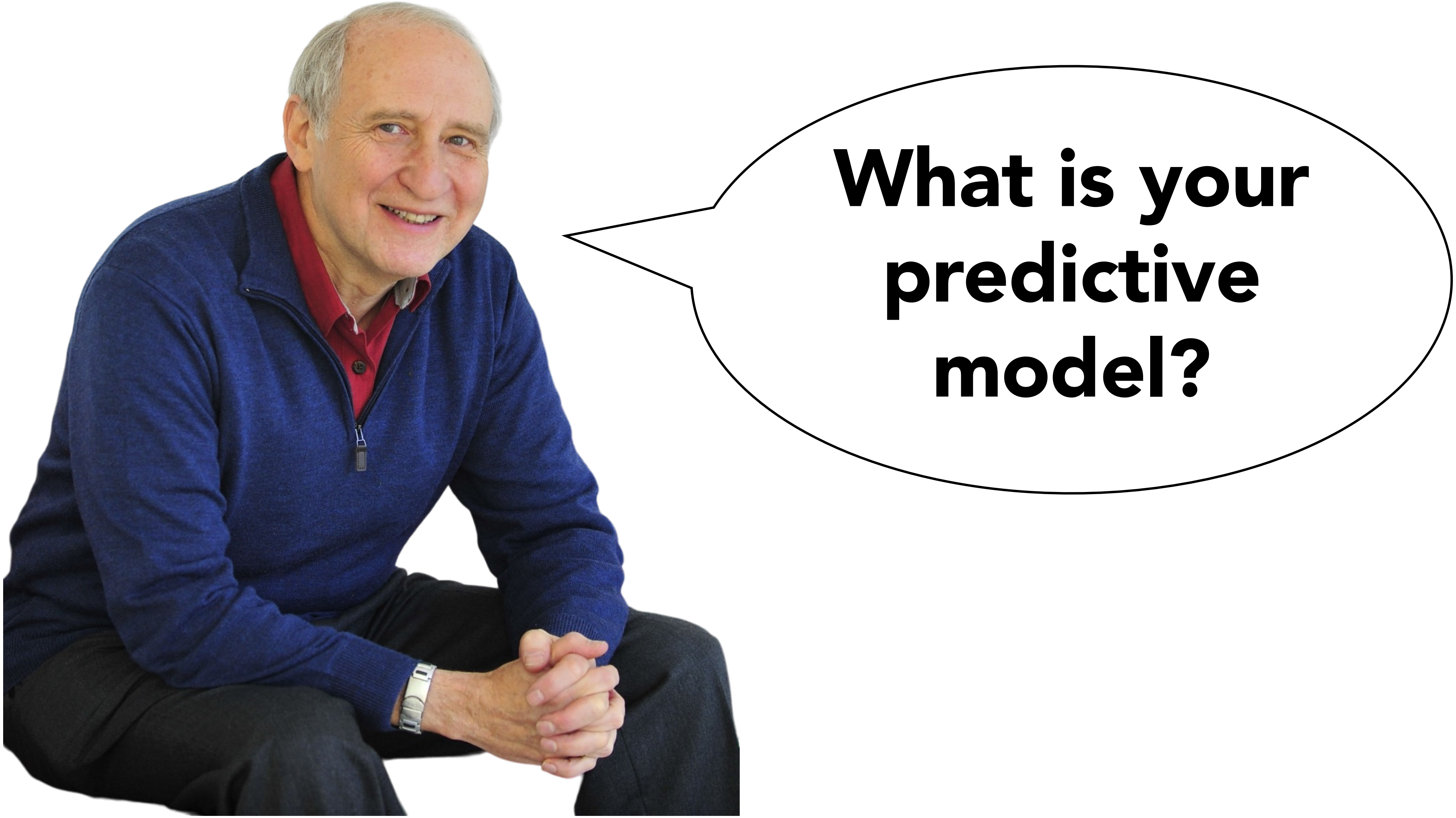
The Theory Crisis in Human-Computer Interaction

Kasper Hornbæk, Professor and Villum Investigator, PhD
University of Copenhagen
kash@di.ku.dk, @khornbaek

UNIVERSITY OF COPENHAGEN

Ben Shneiderman

Ben Shneiderman



Outline of my argument

1. Theory is considered important in HCI
2. The actual role of theory is underwhelming
3. Therefore, we do not reap the benefits that theory promises
4. We should do something about it

1. Theory is considered important in HCI

Examples

Shneiderman & Plaisant (2005) wrote that

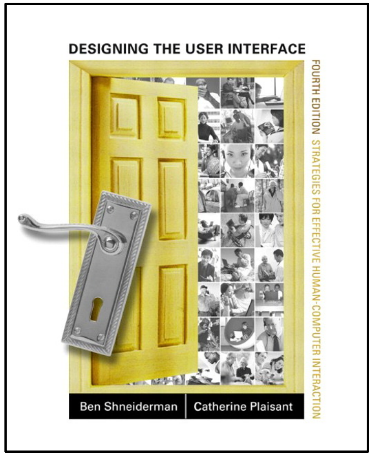
> "One goal for the discipline of human-computer interaction is to go beyond the specifics of guidelines and build on the breadth of principles to develop tested, reliable, and broadly useful theories" (p. 82)

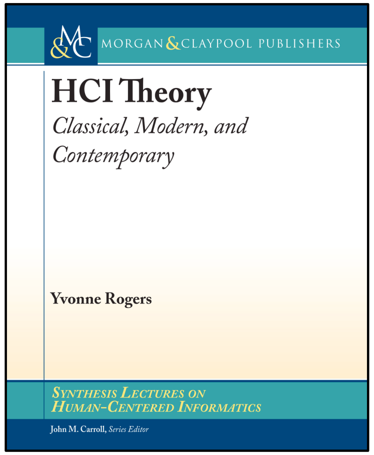
CHI 2027 call for papers

> "Papers […] broadly influence the development of HCI theory, methods, and practice"

Highly cited books on theory (e.g., Rogers 2012)

One of the seven types of contributions in HCI (Wobbrock & Kientz, 2016)

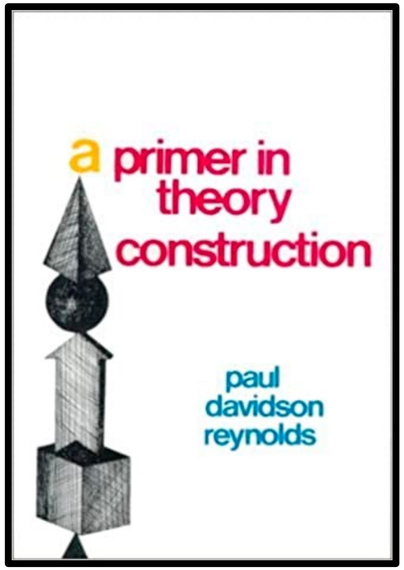




What is theory, anyway?

A propositional view of theory (e.g., Reynolds 1971)

- A tool of the mind to help order and make sense of the world
- Comprises a set of constructs, statements that relate them, mechanisms underlying those statements, and some boundaries of applicability
- Must have empirical content
- Applicable across situations



The simple version (Whetten, 1989)

- What, how, why, who/where/when

Whetten (1989), What Constitutes a Theoretical Contribution, *Academy of Management Review*

What is a theory in HCI?

At CHI, frequently used theories include

- Self-determination theory
- Feminist theories
- Critical race theory
- Activity theory
- Behavior change theory
- Theory of planned behavior
- Soma aesthetic design
- Flow theory

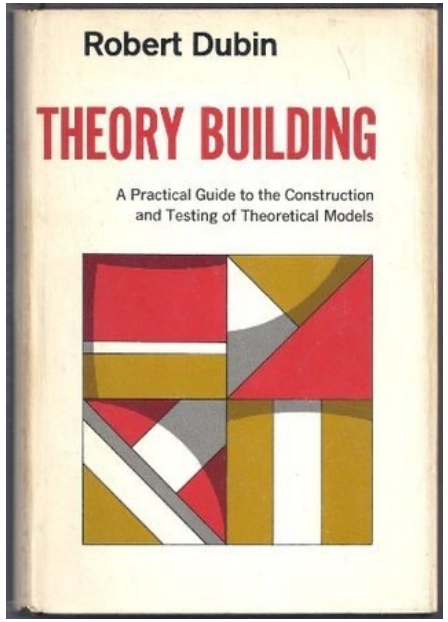
What functions does theory play?

Theory can be used to

- Shape what to value and what to study
- Provide the background for a study or its hypotheses
- Inform methodology
- Design user interfaces
- Analyze data
- Discuss findings

In general, we can separate (Dubin, 1978)

- **understanding** (e.g., explanation, mechanisms) and
- **prediction** (e.g., counter-factual reasoning, control)



2. The role of theory is underwhelming

Data from CHI 2021

743 papers were published at CHI 2021; of those, 283 used some form of theory

Searching for “theo*” in full text and classifying occurrences as well as the papers

Manual coding over the past 5 years

Theory use as background

“[w]e draw from Interaction Design theory to consider designing as a process that does not end with the designers themselves”

“As with others in HCI and design, we have been reading from care ethics [11, 89, 90], posthumanism [45, 46, 89, 103, 104, 108], and ecofeminism [111, 112], finding in them ideas and provocations that seem to suggest an alternative to anthropocentric functionalism”

Theory use as background

"[w]e draw from Interaction Design theory to consider designing as a process that does not end with the designers themselves"

"As with others in HCI and design, we have been reading from care ethics [11, 89, 90], posthumanism [45, 46, 89, 103, 104, 108], and ecofeminism [111, 112], finding in them ideas and provocations that seem to suggest an alternative to anthropocentric functionalism"

Theoretical power posing

This use of theory signals position and intellectual heritage

Can be associated with a soldier mindset (Galef 2021)

Theory for/in discussion

"This might be unsurprising given that the Theory of Planned Behaviour lists social norms as one of three main factors in decision-making"

"This is consistent with Social Penetration Theory, which argues that self-disclosure is the mechanism through which relationships are initiated and strengthened"

Theory for/in discussion

"This might be unsurprising given that the Theory of Planned Behaviour lists social norms as one of three main factors in decision-making"

"This is consistent with Social Penetration Theory, which argues that self-disclosure is the mechanism through which relationships are initiated and strengthened"

Putting a hat on a horse

Theory plays a minor role relative to data or is an afterthought in a discussion

No talkback to theory; no predictions; no up-front work with theory

Strunk & White (1959) *The Elements of Style*

This is a bigger issue in HCI

Limited engagement with theory in HCI (Marshall et al. 2017)

- “We present a citation context analysis of over 3000 citations from 69 papers at CHI2016, which demonstrates that only 4.8% of papers cited are presented as anything other than uncontested fact. In 43% of CHI papers sampled, we found no evidence of any critical engagement.”

Self-determination theory (Tyack and Mekler, 2020, 2024)

- Lack of use of mini-theories

Feminist HCI (Chivukula & Gray, 2020):

- “This paper was mostly given 'credit,' and most frequently 'signposted' to keep readers on track of the topical issues in HCI, with little evidence of explicit use or extension of proposed frameworks”

Theory for design

”we include this functionality because of [X]”

“This simplified animation preview design is inspired by Tufte's minimalism theory for effective information visualization”

Theory for design

"we include this functionality because of [X]"

"This simplified animation preview design is inspired by Tufte's minimalism theory for effective information visualization"

A more general pattern

Among the CHI 2017 best papers, 17 were about construction and design (Oulasvirta & Hornbæk, 2022)

- Nine only mention theory in related work
- Two of the remaining eight engage in more depth with theory
  - “the design of Para was partially based on the theory that a direct-manipulation procedural tool would be compatible with manual practice and enable manual artists to leverage existing skills” (Jacobs et al. 2017)
  - “devise and evaluate several elicitation techniques for information visualization with associated hypotheses” (Kim et al. 2017)

But what about HCI-specific theory?

Theory on usability?

Current literature contains mainly taxonomies of dimensions of usability (e.g., ISO 9241, Nielsen 1993)

Tractinsky (2018) asked if the usability construct is “a dead end”

> “Usability is an umbrella construct. Umbrella constructs are prevalent in scientific fields that are broad, diverse, and lack a unifying research paradigm. Accordingly, umbrella constructs, such as usability, tend to be vague and loose, characteristics that challenge our ability to accumulate and communicate knowledge and to capture real-world phenomena”

Tractinsky (2017), “The Usability Construct: A Dead End?”, *Human-Computer Interaction*

Theory on interaction?

“What do we mean by human-computer interaction? Webster defines ‘interaction’ as ‘mutual or reciprocal action or influence.’ Clearly, humans act on computers and computers influence humans. But how? In what dimensions?”, Winograd (1990)

”The term interaction is field-defining, yet surprisingly confused”, Hornbæk & Oulasvirta (2017)

Hornbæk & Oulasvirta (2017), “What is Interaction?”, *CHI*

3. We do not reap the benefits that theory promises

Benefits that we miss

- Robust generalizations
  - Few replications (2% in Hornbæk et al. 2014)
  - A lot of “it depends” in HCI
- Cumulation of insights across technologies
  - “A big hole in Human-Computer Interaction research”, Kostakos (2015)
- Few insights about interactive behavior packaged for other fields
- Limited development and testing of HCI specific theory

Hornbæk et al. (2014) Is once enough? On the extent and content of replications in human-computer interaction, *CHI*

A theory crisis

1. Theory is considered important in HCI
2. The actual role of theory is underwhelming
3. Therefore, we do not reap the benefits that theory promises

4. We should do something about it

The bigger picture

From phenomena to methods to theory

The turn to theory calls for a research program that focuses on

- How to build theory
- How to use theory
- How to design from theory
- How to create more theory on HCI-specific phenomena like interaction, human-centeredness, user experience, appropriation, human-AI partnerships, and more

VILLUM FONDEN

The TRACTION center

Villum Investigator (2025-2031) 4M€ grant for developing and applying theory about human-computer interaction

The smaller picture

Work deeply with theory across all phases of research

What you can do in any research project:

a. Use theory to plan research
b. Use research to test theory
c. Report with clarity and courage
d. Create theory and implications for it
e. Treat theory as a tool

A. Use theory to plan research

Reason with theory before anything else

What do we already understand?

What can we reasonably predict?

What is your predictive model?

How do you explain this interactive behavior?

What is the mechanism that enables this UI?

A. Use theory to plan research

Form expectations **before** analysis

When we add theory post hoc, we are essentially doing a variant of hypothesizing-after-results-are-known (HARKing, Rubin 2017)

Consider fixing those expectations similarly to preregistration (see Yu et al. 2025 for a worked example)

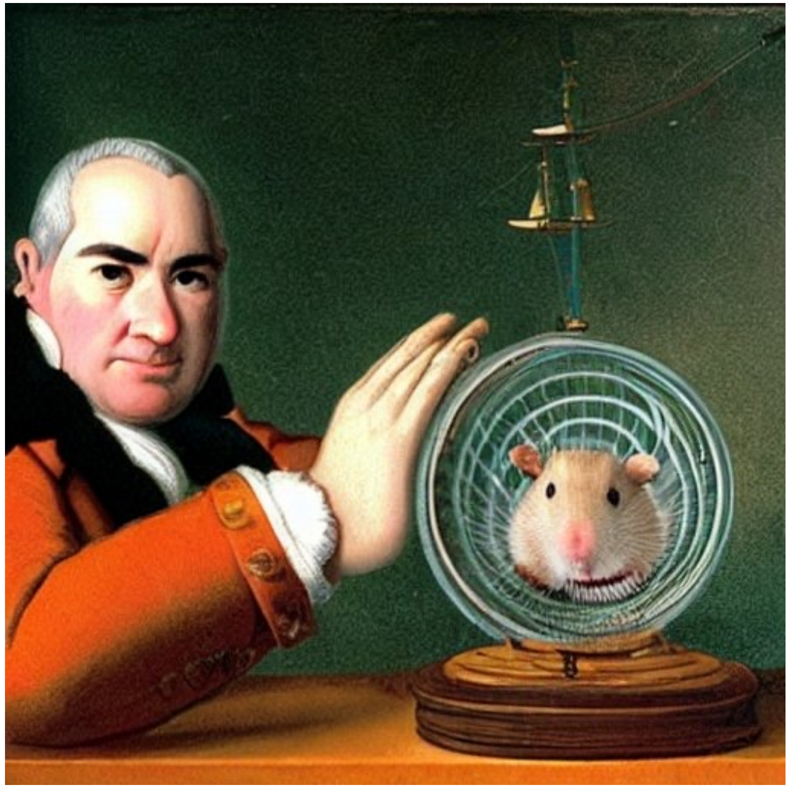
Do not rely only on iteration and empirical studies to resolve theoretical questions

Yu et al. (2025), Deriving Selection Techniques for GUIs based on the Multiple Process Model, *CHI*

A. The inductive hamster wheel

No existing generalization or discovery applies to new situations

Therefore, we use grounded theory, inductive/bottom-up/exploratory studies, or the like



A. All things good come to those who iterate

"Let's get real" (Landauer 1991) and rely on empirical measures and iteration

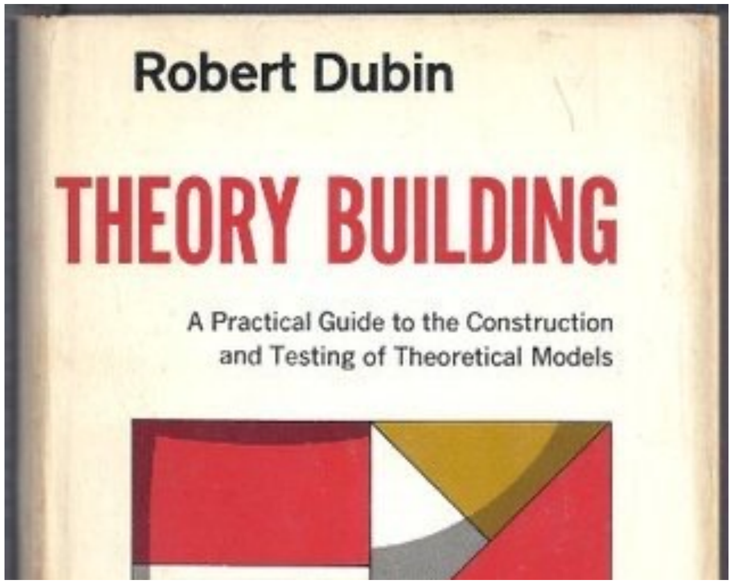
Why bother with theory when we can just A/B test?

Instead, focus on whys and mechanisms (not just on whether something works)

B. Use research to test theory

The job of a researcher is to test theory (Dubin 1978)

- "theorizing is an integral part of empirical investigation, just as empirical analysis has meaning only by reference to a theory from which it is generated"



B. Use research to test theory

Test and talk back to theory

- Make the claims that underlie a design or analysis explicit
- Return to those claims after design or empirical work

Don't pose or position with theory

Use the full range of reasoning capacities in theory

- Beyond taxonomies (e.g., Tyack and Mekler 2020) and measurement tools (e.g., Ekkekakis 2013)

B. Strong inference as a template

Strong inference or forking paths metaphor (Platt, 1964)

- Devise alternative hypotheses
- Design a crucial experiment which will rule out some hypotheses
- Carry out the experiment

B. Focus on conceptual problems

Examples in the literature of conceptual problem solving (Oulasvirta & Hornbæk, 2016):

- Implausible, inconsistent or incompatible theories, frameworks or models
- For example, Borst et al. (2015) aimed to reconcile earlier findings and anomalies around data on interruptions

Oulasvirta & Hornbæk, 2016, "HCI as Problem Solving", *CHI*

B. Good example

“Input mechanisms can produce noisy signals that computers must interpret, and this interpretation can misconstrue the user’s intention. Researchers have studied how interpretation errors can affect users’ task performance, but little is known about how these errors affect learning, and whether they help or hinder the transition to expertise.

B. Good example

"Input mechanisms can produce noisy signals that computers must interpret, and this interpretation can misconstrue the user's intention. Researchers have studied how interpretation errors can affect users' task performance, but little is known about how these errors affect learning, and whether they help or hinder the transition to expertise.

Previous findings suggest that increasing the user's attention can facilitate learning, so frequent interpretation errors may increase attention and learning; alternatively, however, interpretation errors may negatively interfere with skill development."

C. Report with clarity and courage

Dare to make claims and be wrong in predictions

- Most findings in HCI are not even wrong (cf Scheel 2022)
- Avoid ambiguity

A fundamental function of theory is to communicate

- Guess and hypothesize with theory
- Note mistakes and mismatches

C. Report with clarity and courage

Make it possible to be wrong (Hornbæk 2015)

- Is it possible to set up empirical work so that it can be wrong and contradict a theory?
- What would it take for you to change your mind?
- If you cannot, do you hold a theory?

Hornbæk (2015), We must be more wrong in HCI. *interactions*

D. Create theory and implications for it

In addition to testing theory (Dubin), you should also create it

Theory building with a lowercase *t*

Use Whetten's (1989) *W*s

Try to generalize, abstract, and cumulate

What
How
Why
Who/when/where

Let us tackle the key issues in interactive behavior

Usability, user experience, interaction,
reality-based interaction, appropriation, and so on

Implications for Design

Paul Dourish
Department of Informatics
Donald Bren School of Information and Computer Sciences
University of California, Irvine
Irvine, CA 92697-3440 USA
jpd@ics.uci.edu

ABSTRACT
Although ethnography has become a common approach in HCI research and design, considerable confusion still attends both ethnographic practice and the criteria by which it should be evaluated in HCI. Often, ethnography is seen as an approach to field investigation that can generate requirements for systems development; by that token, the major evaluative criterion for an ethnographic study is the implications it can provide for design. Exploring the nature

This process can be seen at work in the research papers produced in a field. Bazerman [3] has detailed the ways in which transformations in the structure and tone of scientific publishing accompanied the transformation of the conduct of science itself, reflecting its increasing professionalization; the process of ensuring conformance to documentary standards is part of the "boundary work" by which disciplinary boundaries are maintained, and even the boundary between "science" and "non-science" is sustained

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, liber fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin ferment Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, ele nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacin Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pe massa.

8 IMPLICATIONS FOR THEORY

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, ad Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, mag augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac tur ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellu tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curab nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, dian

D. Create theory and implications for it

What might implications for theory be?

- The theory itself
- An issue with what Lakatos calls the *protective belt*
- Concerns about a construct
- A new boundary condition
- A suggestion for a more precise mechanism

One of seven kinds of implications in HCI (van Berkel & Hornbæk, 2023)

van Berkel & Hornbæk (2023), "Implications of HCI Research", *interactions*, August 2023, 50-55

E. Treat theory as a tool

Methodological pluralism is well understood (McGrath 1995)

E. The best paper I know on methods

METHODOLOGY MATTERS:
DOING RESEARCH
IN THE BEHAVIORAL
and SOCIAL SCIENCES

JOSEPH E. MCGRATH
PSYCHOLOGY, UNIVERSITY OF ILLINOIS, URBANA

23 OCTOBER 1994

McGrath, J. E. (1995). Methodology matters In *Readings in Human–Computer Interaction* (pp. 152-169)

E. Treat theory as a tool

Methodological pluralism is well understood (McGrath 1995)

Theories are tools for making sense of the world; how could just one do all the work?

- Mix and match, as theories differ in local/global, soc/individual, time-scales, and much more
- Compare theories (e.g., Baumer & Tomlinson 2011)

Don't identify with a theory

- Be able to walk away ("A theory which cannot be mortally endangered cannot be alive", see Platt 1964)

E. Treat theory as a tool (for others)

Theories can be like toothbrushes: Everyone has their own; few want to use someone else's

Fang et al. (2026) looked at a decade of CHI papers

> "Enthusiasm for proposing new frameworks exceeds willingness to iterate on existing ones"

Function	Count
Mention	55
Use	152
Adapt	44
Validate	24
Review	4
Create	336

Fang, Yatani & Hornbæk (2026), What We Talk About When We Talk About Frameworks in HCI, *CHI*

BONUS: Curveballs for CompInt

Theory should both help predict and understand; balance them

Tackle experiences as objective functions and outcomes

Sometimes variance theories and process theories are separated (Mohr, 1982); do quantitative process theories

Many robustness challenges: reproducibility (Iarygina et al. 2026), LLM harking (Kosch & Feger 2026), many modelers (van Dongen et al. 2025)

Automated (re)discovery of theories and models

Large-scale specification, comparison, and simplification of alternative theories and models

Iarygina et al. (2026), On the Computational Reproducibility of Human-Computer Interaction, *CHI*

The Theory Crisis in HCI

1. Theory is considered important in HCI
2. The actual role of theory is underwhelming
3. Therefore, we do not reap the benefits that theory promises
4. We should do something about it

Kasper Hornbæk, University of Copenhagen
kash@di.ku.dk. @khornbaek

References

Baumer, E. P. S., & Tomlinson, B. (2011). Comparing activity theory with distributed cognition for video analysis: Beyond “kicking the tires.” Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, 133–142.

Borst, J. P., Taatgen, N. A., & van Rijn, H. (2015). What makes interruptions disruptive? A process-model account of the effects of the problem state bottleneck on task interruption and resumption. CHI 2015, , 2971–2980.

Chivukula & Gray (2020). Bardzell's "Feminist HCI" Legacy: Analyzing Citational Patterns. In Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems (CHI EA '20).

Dourish, P. (2006), “Implications for design”, *CHI* 2006

Ekkekakis, P. (2013). The measurement of affect, mood, and emotion: A guide for health-behavioral research. Cambridge University Press.

Fang, S., Yatani, K., & Hornbæk, K. (2026). What we talk about when we talk about frameworks in HCI. Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems

Dubin, R. (1978), *Theory Building*.

Galef, J. (2021). *The Scout Mindset: Why Some People See Things Clearly and Others Don't*. Penguin.

Gioia, D. A., & Pitre, E. (1990). Multiparadigm perspectives on theory building. *Academy of management review*, *15*(4), 584-602.

Gugenheimer, J., Stemasov, E., Frommel, J., & Rukzio, E. (2017, May). Sharevr: Enabling co-located experiences for virtual reality between hmd and non-hmd users. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (pp. 4021-4033).

Hornbæk, K. (2015). We must be more wrong in HCI research. Interactions, 22(6), 20–21.

Hornbæk & Oulasvirta (2017), “What is Interaction?”, CHI 2017

Hornbæk, K., Sander, S. S., Bargas-Avila, J. A., & Grue Simonsen, J. (2014, April). Is once enough? On the extent and content of replications in human-computer interaction. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 3523-3532).

Iarygina, O., Hornbæk, K., & Mottelson, A. (2026). On the computational reproducibility of human-computer interaction. *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems.*

References, cont’d

Jacobs, J., Gogia, S., Měch, R., & Brandt, J. R. (2017, May). Supporting expressive procedural art creation through direct manipulation. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (pp. 6330-6341).

Kim, Y. S., Reinecke, K., & Hullman, J. (2017, May). Explaining the gap: Visualizing one's predictions improves recall and comprehension of data. In Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems (pp. 1375-1386).

Kosch, T., & Feger, S. (2026). Prompt-hacking: The new p-hacking? Communications of the ACM.

Kostakos, V. (2015). The big hole in HCI research. *interactions*, *22*(2), 48-51.

Liu, Y., Li, H., Kostakos, V., Goncalves, J., Hosio, S., & Hu, F. (2014). An empirical investigation of mobile government adoption in rural China: A case study in Zhejiang province. *Government Information Quarterly*, *31*(3), 432-442.

Marshall, J., Linehan, C., Spence, J., & Rennick Egglestone, S. (2017, May). Throwaway citation of prior work creates risk of bad HCI research. CHI 2017 (pp. 827-836)

Mohr, L. B. (1982). Explaining organizational behavior: The limits and possibilities of theory and research. Jossey-Bass.

Nielsen, J. (1993). Usability engineering. Academic Press.

McGrath, J. E. (1995). Methodology matters: Doing research in the behavioral and social sciences. In Readings in Human–Computer Interaction (pp. 152-169). Morgan Kaufmann.

References, cont’d

Oulasvirta, A., & Hornbæk, K. (2022). Counterfactual thinking: What theories do in design. International Journal of Human–Computer Interaction, 38(1), 78-92.

Platt, J. R. (1964). Strong inference. *Science, 146*(3642), 347–353.

Reynolds, P. D. (2015). Primer in theory construction: An A&B classics edition. Routledge.

Rogers, Y. (2012). *HCI theory: Classical, modern, and contemporary*. Morgan & Claypool.

Rubin, M. (2017). When does HARKing hurt? Identifying when different types of undisclosed post hoc hypothesizing harm scientific progress. *Review of General Psychology, 21*, 308–32

Scheel, A. M. (2022). Why most psychological research findings are not even wrong. *Infant and Child Development, 31*, e2295

Shneiderman & Plaisant (2005) . *Designing the User Interface: Strategies for Effective Human-Computer Interaction*, 4th ed.

Strunk, W., Jr., & White, E. B. (1959). The elements of style. Macmillan.

Tractinsky, N. (2018). The usability construct: a dead end?. *Human–Computer Interaction*, *33*(2), 131-177.

van Dongen, N., Finnemann, A., De Ron, J., Tiokhin, L., Wang, S., Algermissen, J., Altmann, E. C., Bahník, Š., Chuang, L.-C., Dumbravă, A., Fünderich, J. H., Geiger, S. J., Gerasimova, D., Golan, A., Herbers, J., Jekel, M., Kunnari, A., Lin, Y.-S., Moreau, D., Oberholzer, Y., et al. (2025). Practicing theory building in a many modelers hackathon: A proof of concept. *Meta-Psychology, 9*.

Whetten, D. A. (1989). What constitutes a theoretical contribution?. *Academy of management review*, *14*(4), 490-495.

Wobbrock , J. & Kientz, J. (2016). Research contributions in human-computer interaction. *interactions* 23, 3), 38–44.

Yu, D., Roberts, J., Hornbæk, K., & Bergström, J. (2025). Deriving selection techniques for GUIs based on the multiple process model. *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 1–16.